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Physics-Aware Multi-Task Learning for Atmospheric Turbulence Parameterization: Auxiliary Tasks versus Architectural Conditioning

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1. Introduction and Motivation

❖ Large Eddy Simulation (LES):

- ❑ Directly **resolves large turbulent eddies**, responsible for most of the turbulent energy and transport.
- ❑ Parameterizes the **small-scale turbulent features** that are primary sources of mixing/dissipation **using subgrid-scale models (**e.g. **Smagorinsky-Lilly)**.

❖ Motivation

- ❑ Traditional turbulence models: computationally expensive; may not be accurate!
- ❑ **Artificial Intelligence (AI)** techniques have shown promise [Maulik et al. (2019), Pal (2020), Kashinath et al. (2021), Subel et al. (2021), Beucler et al. (2023), Guan et al. (2023)].

➢ **Our Approach:**
AI based Emulation of the SGS scheme (Smagorinsky-Lilly)

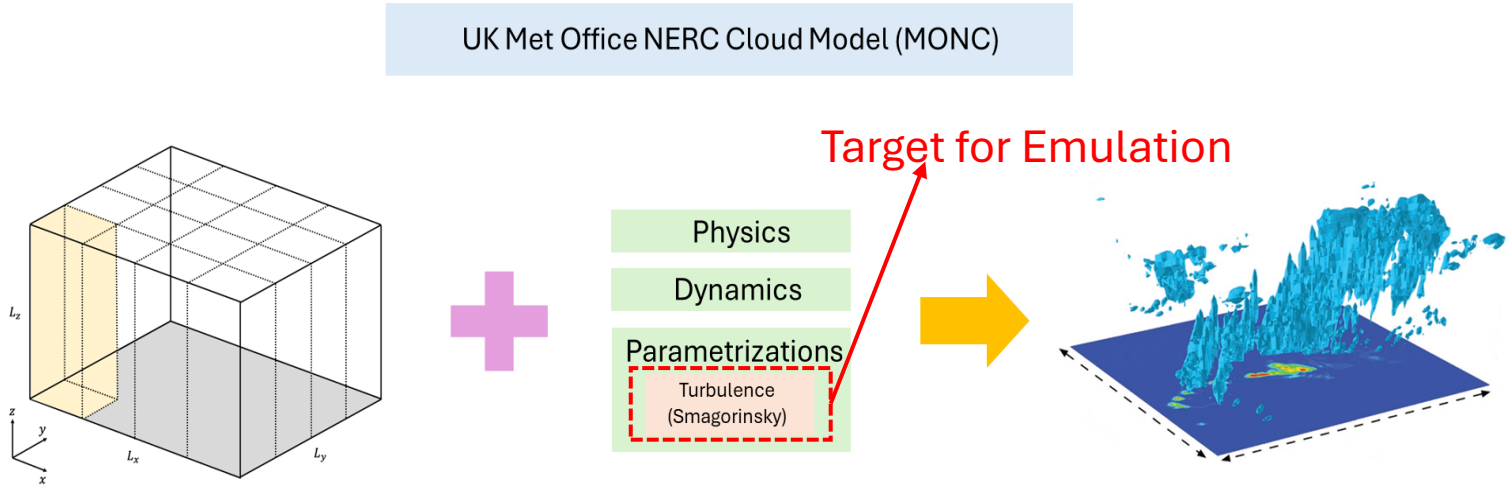


Fig 1: Schematic Representation of the LES setup

1.1. Background: Smagorinsky-Lilly SGS Closure inside MONC

- LES filters resolved motions from the subgrid-scale turbulence.
- MONC's SGS parameterization (Smagorinsky-Lilly closure) computes two coefficients, modulated by stability conditions.

K_m: Eddy Viscosity (m^2/s)

K_h: Eddy Diffusivity (m^2/s)

Richardson Number (Ri): Represents turbulence intensity and stability

$$Ri = \frac{(g/\theta_0)(\partial\bar{\theta}/\partial z)}{\left(\frac{\partial\bar{u}}{\partial z}\right)^2 + \left(\frac{\partial\bar{v}}{\partial z}\right)^2}$$

Ri discriminates the turbulence regime

Ri < 0 : Unstable
Buoyancy-driven turbulence

0 < Ri < 0.25 : Stable
Mechanical mixing under stratification

Ri > 0.25 : Strongly Stable
Turbulence supressed entirely

Note: Here g is gravitational acceleration, θ_0 is a reference potential temperature, z is the vertical coordinate, and $(\bar{u}, \bar{v})$ are the filtered horizontal velocity components

2. Research Questions for this study

- **RQ1: Multi-resolution training from sparse points?**
  Is multi-resolution training and full-3D inference feasible from sparsely sampled, stencil-based point datasets covering a small fraction of the domain?

- **RQ2: Physics-guided vs. data-driven, at equal data?**
  Do physics-guided frameworks outperform implicit / minimally-guided data-driven architectures given the same training data?

- **RQ3: Should Physics persist at inference, or train-time only?**
  Should physics knowledge persist at inference through architectural conditioning, or act only during training via gradient regularization?

- **RQ4: Is MTL the right choice, and do CV weightings transfer?**
  Is multi-task learning (MTL) a good fit, and do automated task-weighting methods transfer from computer vision to atmospheric turbulence modeling?

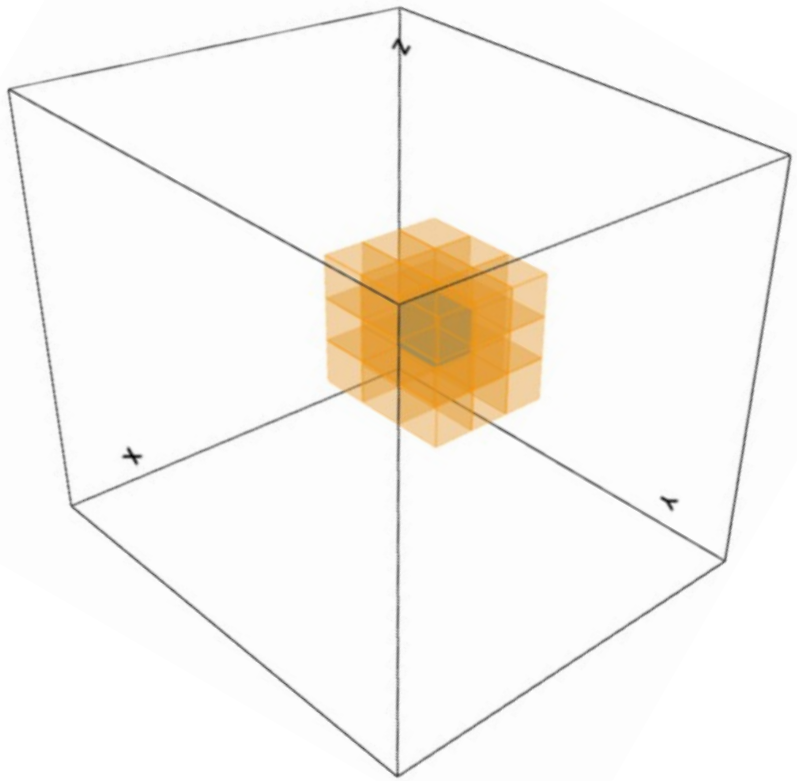
- **RQ5: Does architectural complexity help or hinder?**
  Does architectural complexity improve or hinder generalization across regimes with different turbulence characteristics?

Note: The results in the paper are based on offline training and inference. Ongoing work deals with coupled Physics+ML online inference (simulations)

3. Data and Methodology

Point-wise stratified sampling and Multi-task Learning (MTL) framework:

Sampling: Regime based selection of pixels/grid points with chosen (star) stencil



+

MLP/Res-MLP/TabTransformer based models

- **Input Features:** 54 (Including neighbourhood pixels, normalized coordinates and others)
- **Feature Scaling:** StandardScaler
- **Sampling:** Full Domain – Edges (2 pixels: Halo Cells)
- **Regime based data balance:** Ensured
- **Simulation Type:** RCE (Deep Convection: 1km dx/dy) + ARM (Shallow Convection: 50m dx/dy)

Training: 4.05 M samples (66% RCE, 34 % ARM); ~5% of available data
Evaluation/Testing: 202.8 M samples (RCE+ARM)

3.1. Training Data Composition

- Radiative Convective Equilibrium (RCE): Representing **deep tropical convection** with vigorous updrafts and organized convective structures
  - Spatial resolution: $\boldsymbol{\Delta x = \Delta y = 1\ km}$
  - Domain: 64 x 64 x 99
  - Vertical layers: 99 layers with varying Δz
  - Simulation time steps covered: 200 s to 50,000 s (model time)
  - Sparse sample coverage: **24.4 %** of full 3D domain
  - Contribution to training data mix: **66 %**

- Atmospheric Radiation Measurement (ARM): Representing **shallow mid-latitude cumulus** clouds with weaker, more distributed turbulence and different thermodynamic profiles
  - Spatial resolution: $\boldsymbol{\Delta x = \Delta y = 50\ m}$
  - Domain: 192 x 192 x 220
  - Vertical layers: 220 layers with varying Δz
  - Simulation time steps covered: 10,800 s to 46,800 s (model time)
  - Sparse sample coverage: **1.9 %** of full 3D domain
  - Contribution to training data mix: **34 %**

3.2. ML model architecture: 54 MTL configs (2 physics pathways × 3 architectures × 3 task weighting methods × 3 random seeds)

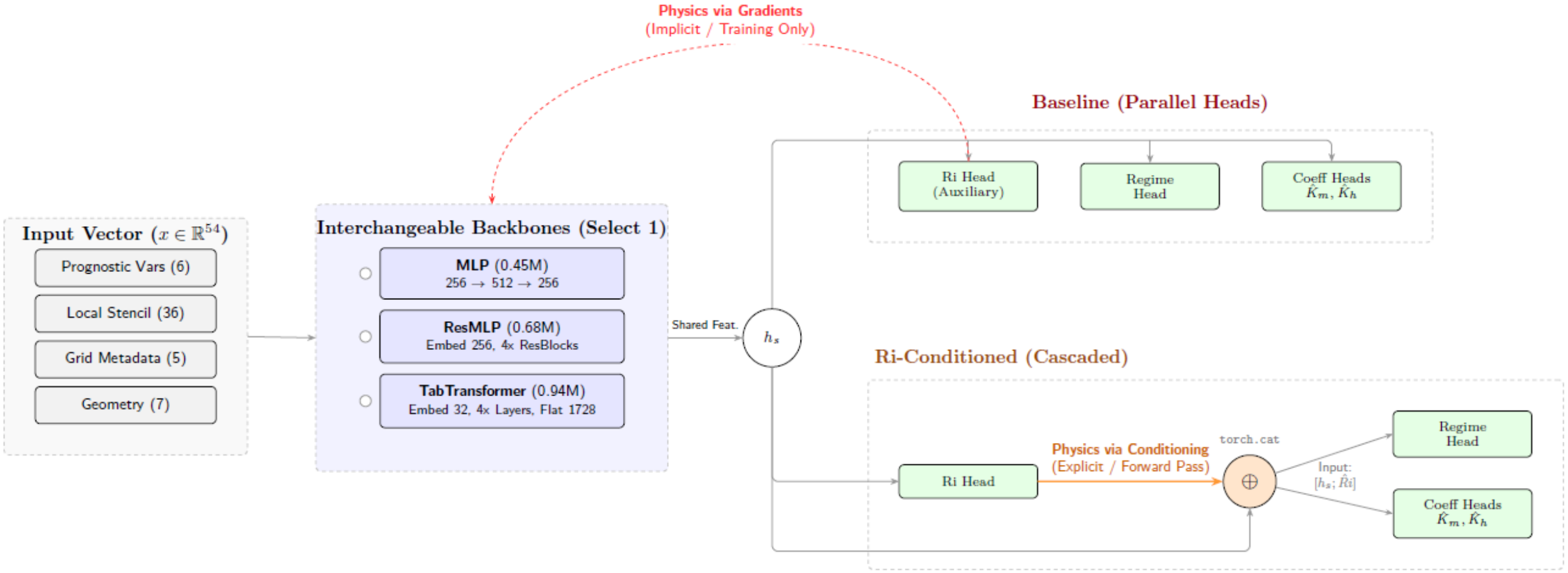


Fig 1: Physics-Aware MTL for SGS Parameterization. Maps atmospheric states to turbulence coefficients (K_m, K_h) across various backbones.

- **(Top) Baseline:** Physics acts as an auxiliary task via gradient regularization (training only).
- **(Bottom) Ri-Conditioned:** Explicit physics integration where predicted Richardson number (Ri) directly conditions the coefficient outputs (training & inference).

3.3. MTL Training set up

Table 1: Neural Network Architecture Specifications

Component	MLP	ResMLP	TabTransformer
Parameters	0.45M	0.68M	0.94M
Backbone layers	[256, 128, 64]	[256, 128, 64]	4 attention layers
Skip connections	None	Identity mapping	None
Attention heads	None	None	4 heads
Normalization	LayerNorm	LayerNorm	LayerNorm
Activation	ReLU	ReLU	GELU
Prediction heads	[32, 16] → 1 (regression), [32, 3] → softmax (regime)		

MTL Training Loss

$$L_{total} = L_{k_m} + L_{k_h} + L_{Ri} + L_{regime}$$

- **Uncertainty weighting** [Kendal et al. (2018)] learns task-specific homoscedastic uncertainty parameters σ_k that automatically balance objectives based on prediction confidence:

$$L_{total} = \sum_k \frac{1}{2\sigma^2} L_k + \log \sigma_k$$

- **Dynamic Weight Averaging (DWA)** [Liu et al. (2019)] adapts weights based on relative loss improvement rates:

$$w_k(t) = \frac{K \exp(r_k(t-1)/T)}{\sum_j \exp(r_j(t-1)/T)} \qquad r_k(t) = \frac{L_k(t)}{L_k(t-1)}$$

4. Results: Task Weighting Experiments

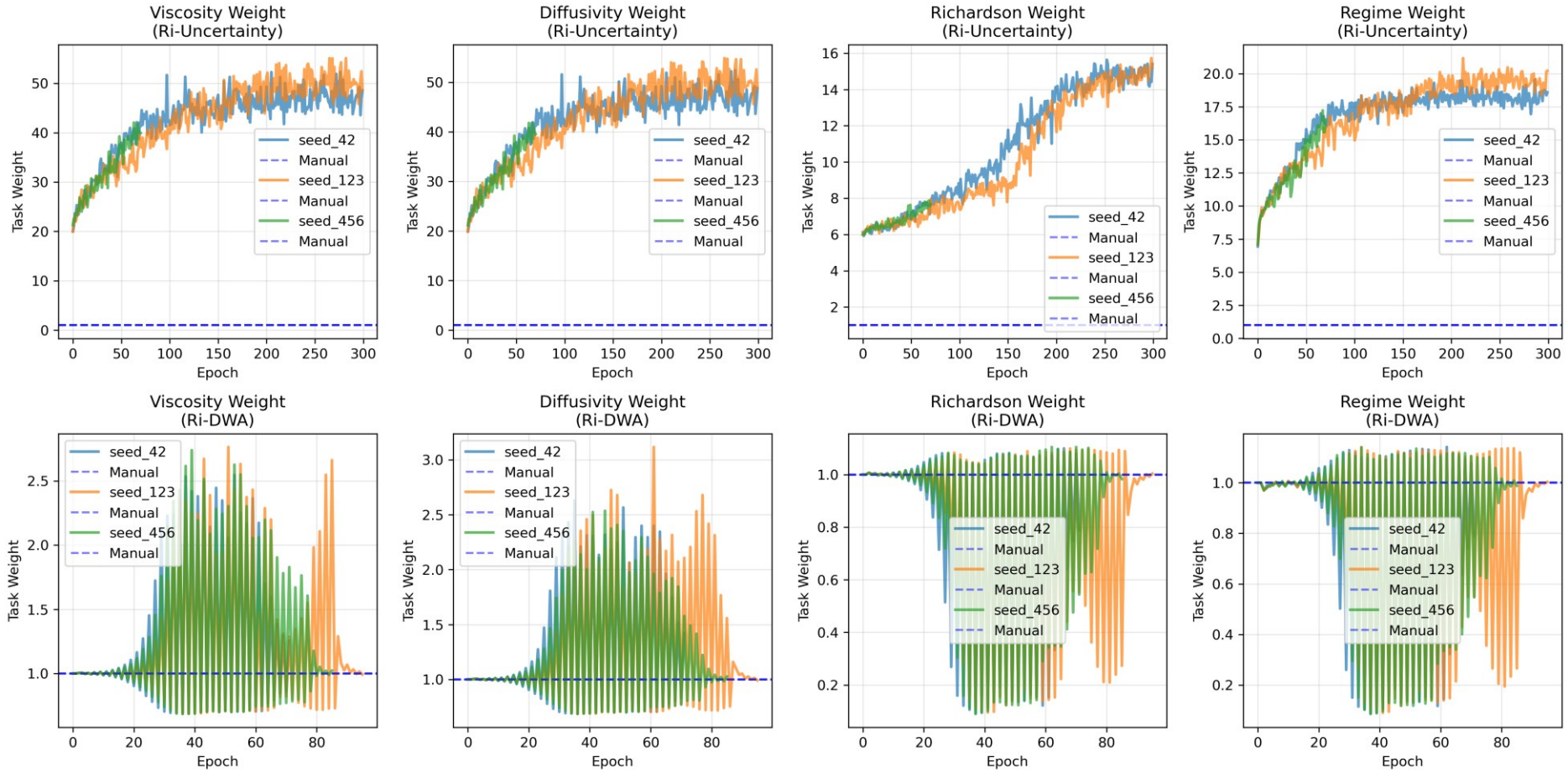


Fig 2: Sample MTL training plots for Ri-MLP architecture:

- **Uncertainty weighting wins** in all 54 configurations (refer Zenodo repo and the paper for all results)
- DWA weighting is oscillatory across all seeds and model configurations.

Table 2: Test-split viscosity R^2 (mean ± std, 3 seeds): Baseline config

Method	MLP	ResMLP	TabTransformer
Manual	0.626 ± .031	0.708 ± .004	0.633 ± .007
Uncertainty	0.835 ± .004	0.869 ± .005	0.821 ± .004
DWA	*0.601 ± .047*	*0.300 ± .013*	*0.258 ± .014*

4.1. Results: Quantitative evaluation on full 3D domain and active turbulence regions

Table 3: Full-Domain RCE Performance (49 timesteps, 19.87M predictions/timestep)

Model	Visc R^2	Diff R^2	Ri R^2	Reg Acc	σ_{R^2}
Baseline Approach (Uncertainty Weighting)					
Baseline-MLP	0.801	0.798	0.569	97.3%	0.064
Baseline-ResMLP	0.782	0.778	0.321	97.6%	0.071
Baseline-TabT	0.713	0.702	0.804	96.8%	0.082
Ri-Conditioned Approach (Uncertainty Weighting)					
Ri-MLP	0.793	0.789	0.601	97.4%	0.089
Ri-ResMLP	0.771	0.766	0.358	97.5%	0.118
Ri-TabT	0.704	0.693	0.789	96.6%	0.095

Table 4: Full-Domain ARM Performance (25 timesteps, 202.75M predictions/timestep)

Model	Visc R^2	Diff R^2	RMSE (Visc)	σ_{R^2}
Baseline Approach (Uncertainty Weighting)				
Baseline-MLP	0.51 ± 0.50	0.61 ± 0.47	0.81	0.50
Baseline-ResMLP	0.33 ± 2.96	0.36 ± 2.74	0.87	2.96
Baseline-TabT	-0.18 ± 1.42	-0.05 ± 1.30	1.14	1.42
Ri-Conditioned Approach (Uncertainty Weighting)				
Ri-MLP	0.57 ± 0.45	0.65 ± 0.42	0.78	0.45
Ri-ResMLP	0.14 ± 4.81	0.12 ± 3.87	1.06	4.81
Ri-TabT	0.10 ± 1.40	0.03 ± 1.25	1.06	1.40

Table 5: Performance on Active Turbulence Regions (coefficient > 0.01)

Regime	Model	Visc R^2	Diff R^2	Var Ratio	%Active
RCE (1.63M active samples of 19.87M total)					
RCE	Baseline-MLP	0.856	0.854	0.753	8.2%
RCE	Ri-MLP	0.824	0.821	0.612	8.2%
RCE	Baseline-ResMLP	0.847	0.844	0.937	8.2%
RCE	Ri-ResMLP	0.833	0.829	0.917	8.2%
ARM (41.4M active samples of 202.75M total)					
ARM	Baseline-MLP	0.096	0.382	1.147	20.4%
ARM	Ri-MLP	0.387	0.554	0.809	20.4%
ARM	Baseline-ResMLP	−2.27	−0.71	3.824	20.4%
ARM	Ri-ResMLP	−9.08	−4.15	10.59	20.4%

- ❖ Data rich regimes (In distribution): Baseline MLP wins
- ❖ Data poor (Out of distribution): Ri-MLP wins

➢ There is merit in the Ri-conditioning approach!

4.2. Results: Distribution comparison across active turbulence regions in RCE

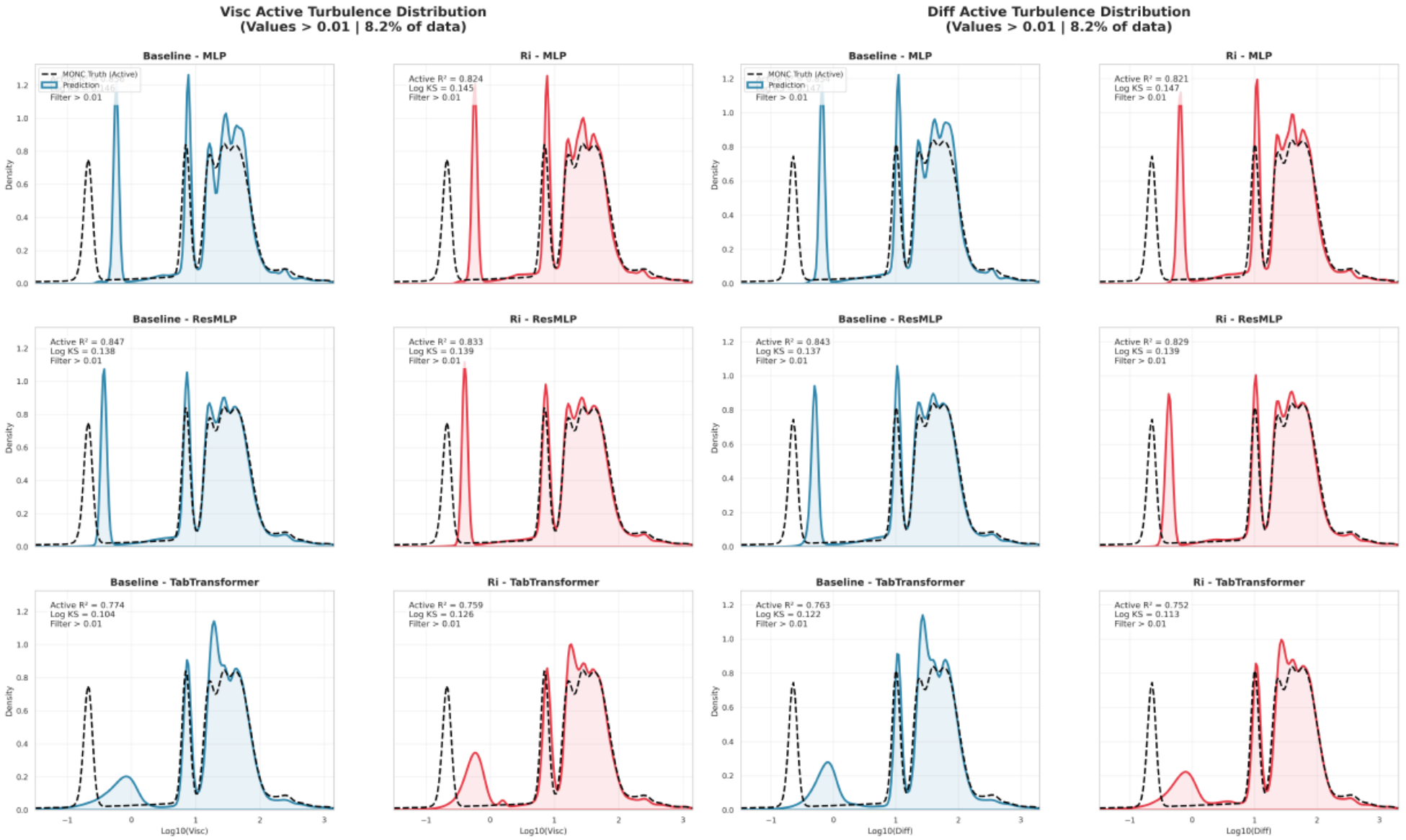


Fig 3: Distribution matching for RCE active turbulence regions: viscosity (left) and diffusivity (right). All ML predictions (coloured) achieve reasonable alignment with truth distributions (black dotted).

4.3. Results: Distribution comparison across active turbulence regions in ARM

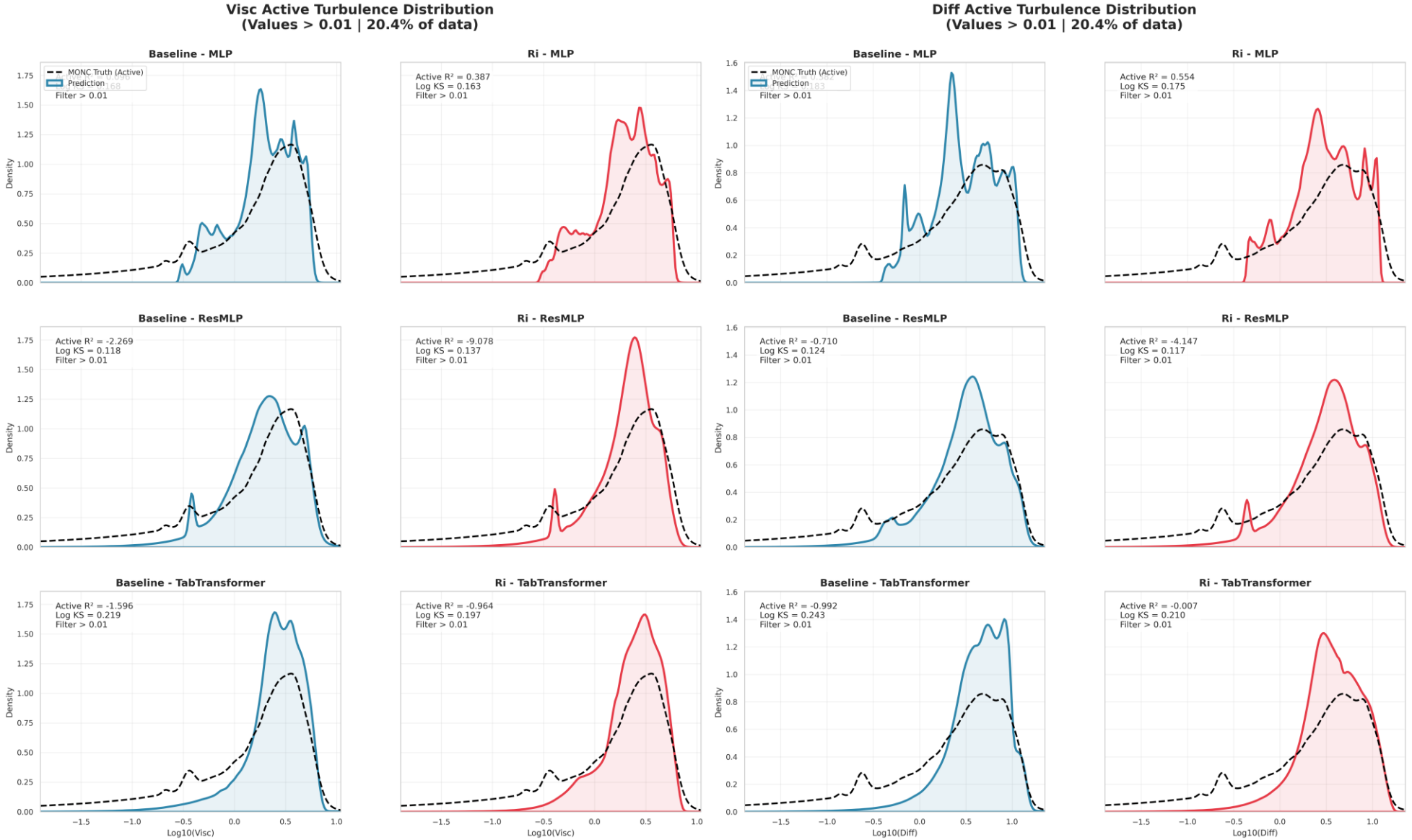


Fig 4: Distribution matching for ARM active turbulence regions: viscosity (left) and diffusivity (right). All ML predictions (coloured) achieve reasonable alignment with truth distributions (black dotted).

4.4. Results: Sample Domain Averaged Vertical Profiles after (offline) inference using Ri-conditioned models

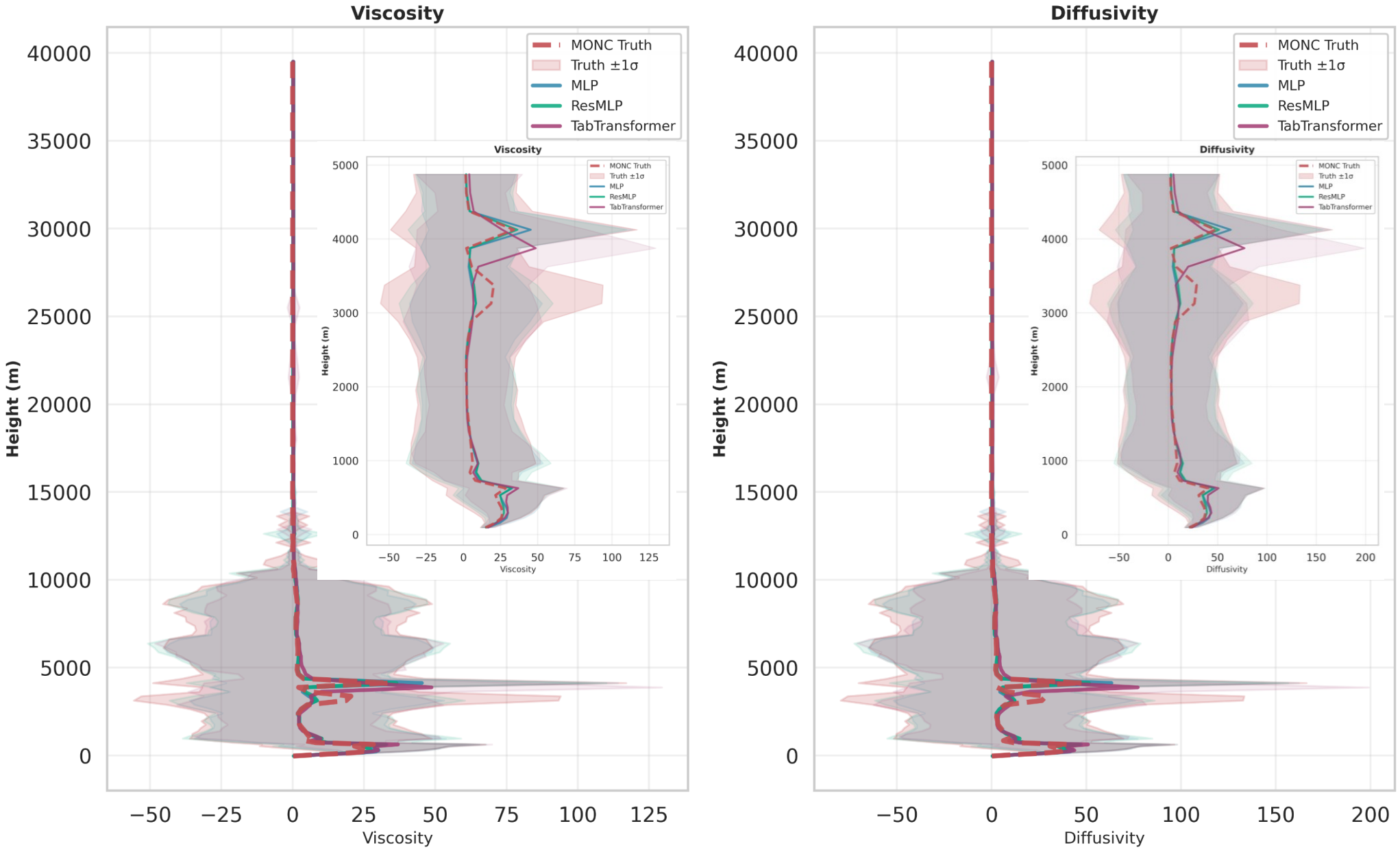
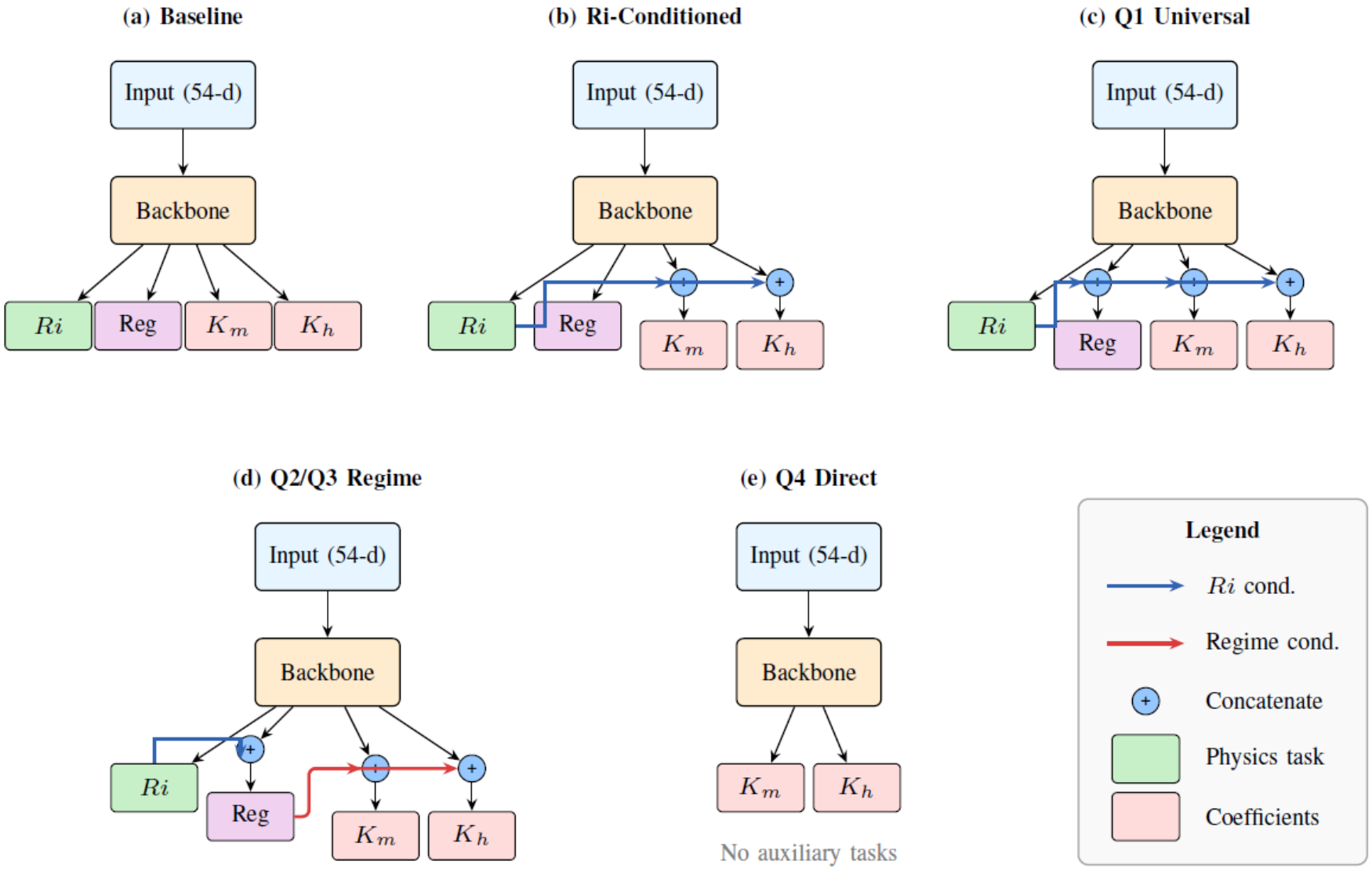


Fig 5: Domain averaged vertical profiles for Ri condition models. Colour Shadings variability across each vertical layer

5. Extended MTL configs (WCCI IJCNN 2026): 54 (6 conditioning × 3 architectures × 3 seeds)



Source: From IEEE WCCI IJCNN 2026 paper (Panda et al. 2026)

Fig 6: Multi-task architectures for SGS coefficient prediction. (a) Baseline: Auxiliary tasks (Ri, Regime) provide gradient regularization only. (b) Ri-Conditioned: Predicted Ri concatenated before coefficient heads. (c) Q1: Ri conditioning applied universally. (d) Q2/Q3: Two-stage cascade with Ri conditioning regime, then regime conditioning coefficients; Q3 uses staged training. (e) Q4: Ablation baseline with coefficient heads only, no auxiliary tasks.

5.1. Few quantitative evaluation metrics from the full domain inference results

R2 scores for MLP variants under RCE (dominant, in-distribution like) and ARM (rare, out-of-distribution like) regimes

Model/R^2 Scores	RCE Visc	RCE Diff	ARM Visc	ARM Diff	Mean Visc	Mean Diff
Q3-MLP	0.811	0.808	0.616	0.701	0.714	0.754
Baseline-MLP	0.801	0.798	0.540	0.644	0.671	0.721
Q4-MLP	0.784	0.782	0.642	0.645	0.713	0.713
Ri-MLP	0.773	0.770	0.670	0.734	0.721	0.752
Q2-MLP	0.772	0.769	0.485	0.647	0.629	0.708
Q1-MLP	0.722	0.721	0.119	0.424	0.421	0.572

- **Ri-MLP on ARM: 0.670 / 0.734 — +24% over Baseline.** Explicit conditioning helps precisely under distribution shift.
- **Q1 collapses (0.119/0.424):** universal conditioning cascades Ri errors into coefficients: targeted beats naive expansion.

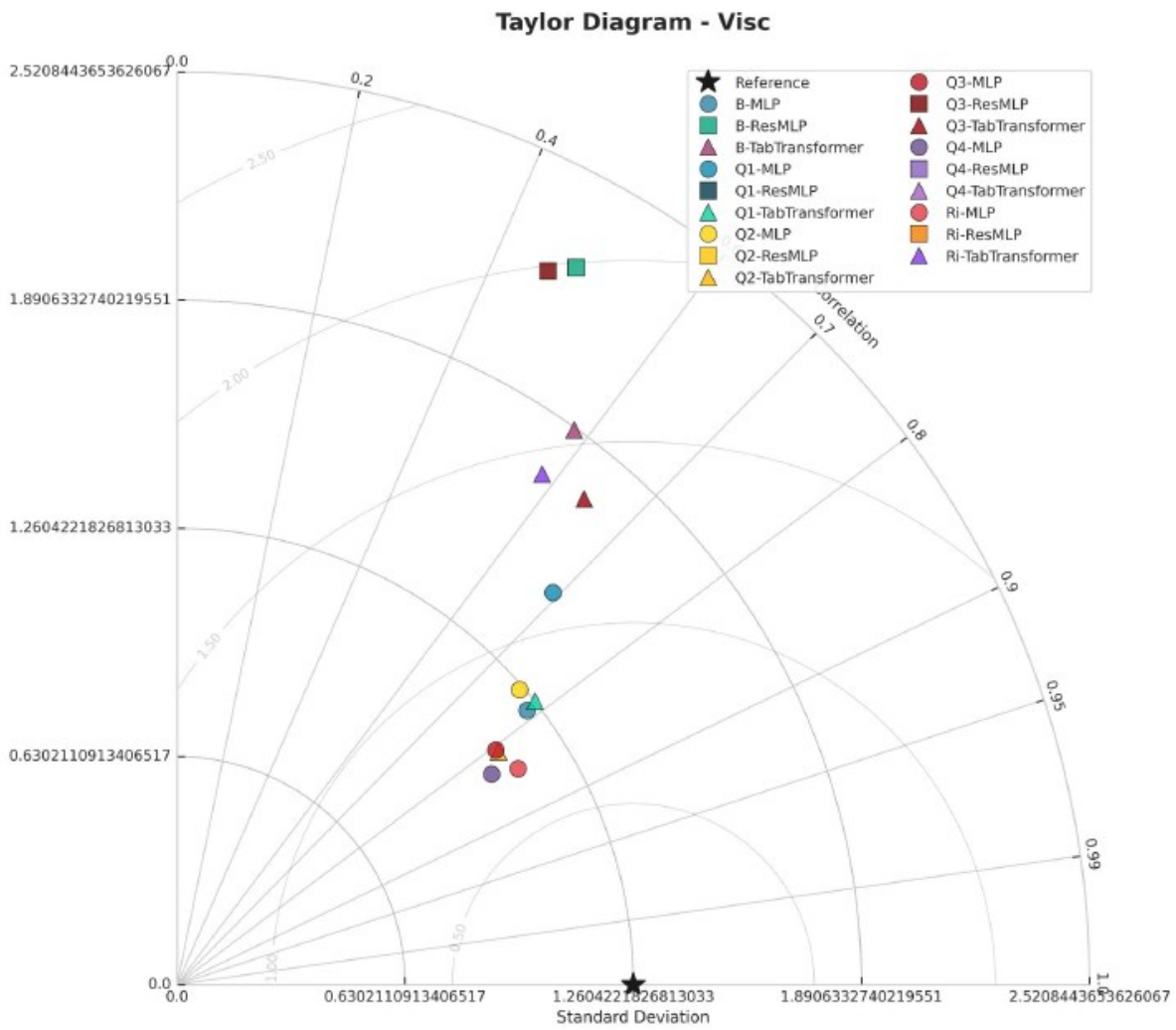
$$KGE = 1 - \sqrt{[(r-1)^2 + (\alpha - 1)^2 + (\beta - 1)^2]}$$

where, r is the correlation coefficient, α is the variability ratio ($\sigma_{pred}/\sigma_{truth}$), β is the bias ratio ($mean_{pred}/mean_{truth}$).

Data · Model	KGE	r	α	β
RCE · Q3-MLP	0.809	0.901	0.905	1.13
ARM · Q3-MLP	0.498	0.805	0.868	1.44
ARM · Ri-MLP	0.426	0.845	0.885	1.54
ARM · Baseline-ResMLP	*-0.025*	*0.487*	*1.80*	*1.39*

- MLPs seem Bias dominated: Could be fixed with data augmentation.
- ResMLPs seem Variance dominated: Need careful constraints on the residual pathways.
- All models could benefit from more training data, as currently the models have learnt the physics well (high correlation, negligible physics violation).

5.2. Taylor Diagram for all ML models: From IEEE WCCI IJCNN 2026 paper (Panda et al. 2026)



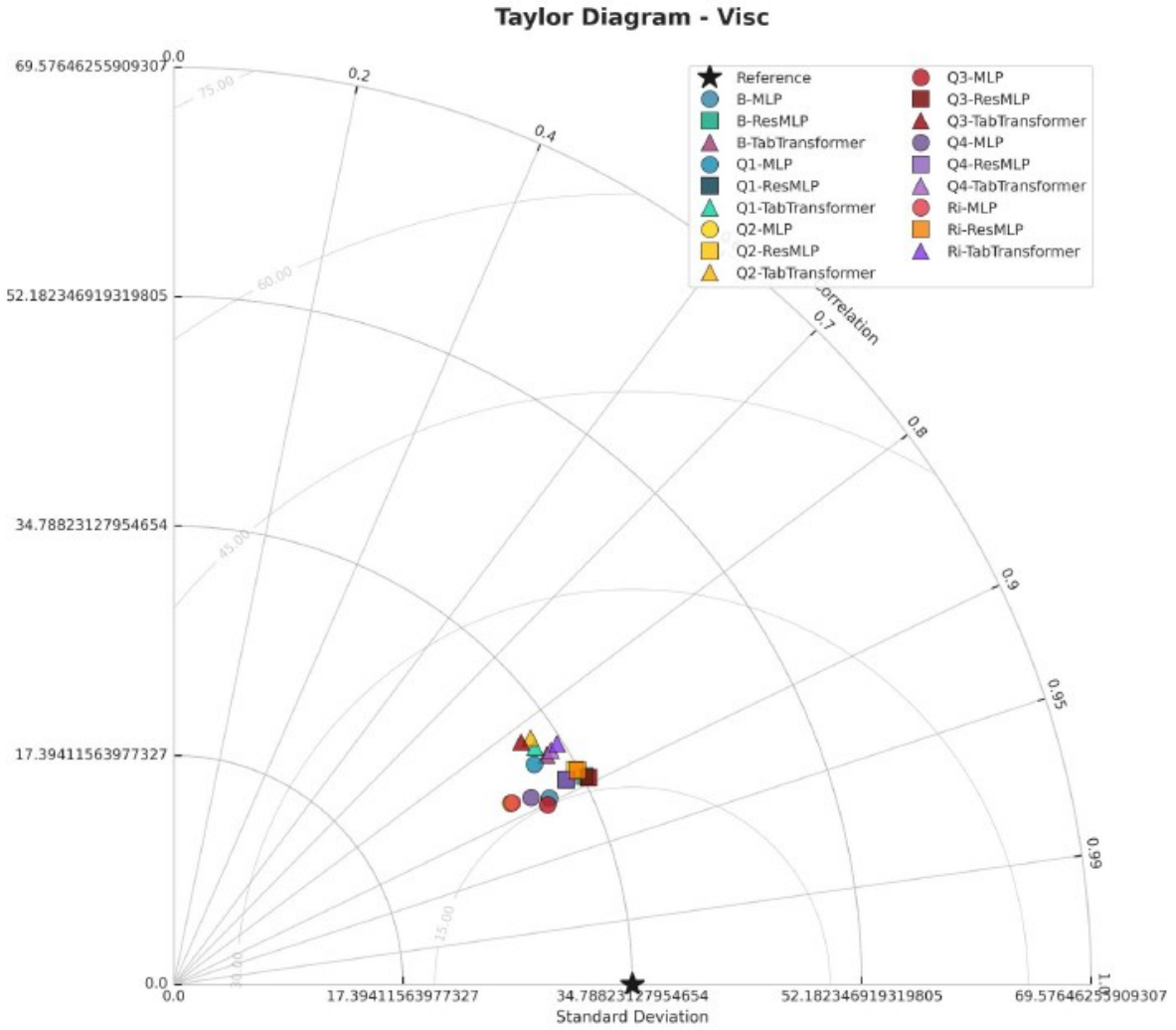
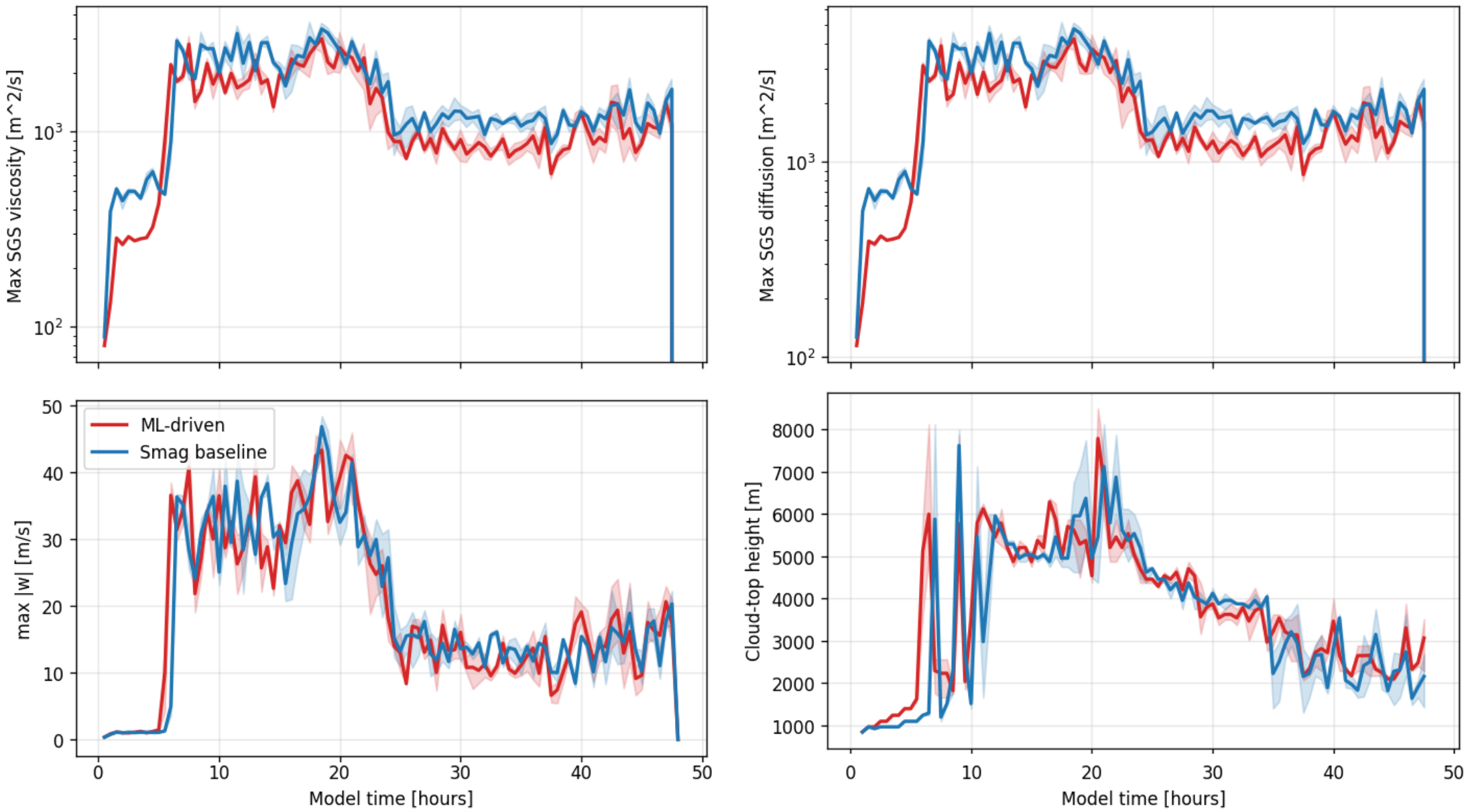


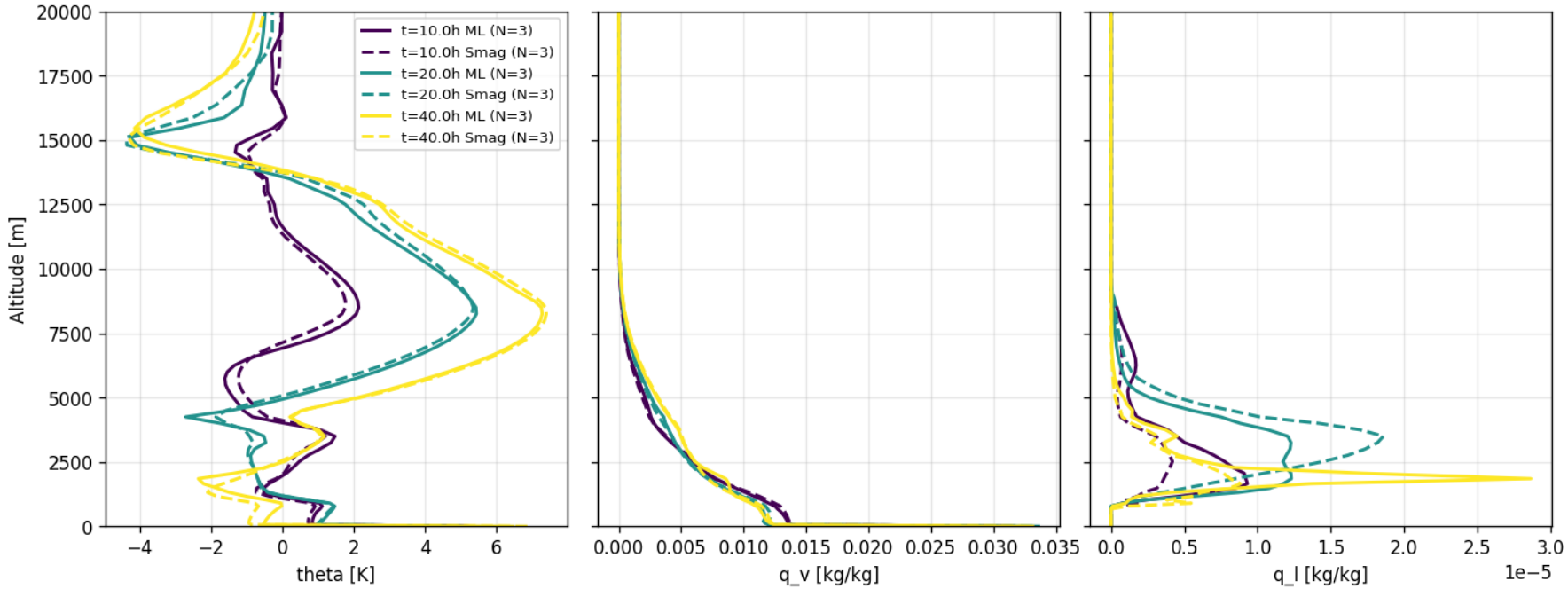
Fig 7: Taylor diagrams comparing viscosity prediction on ARM (left) and RCE (right). While all models show competitive in-distribution accuracy on RCE, only MLP variants (circles) maintain robust variance and correlation under the ARM distribution shift. Complex architectures like ResMLP (triangles) and TabTransformer (squares) exhibit severe variance explosion.

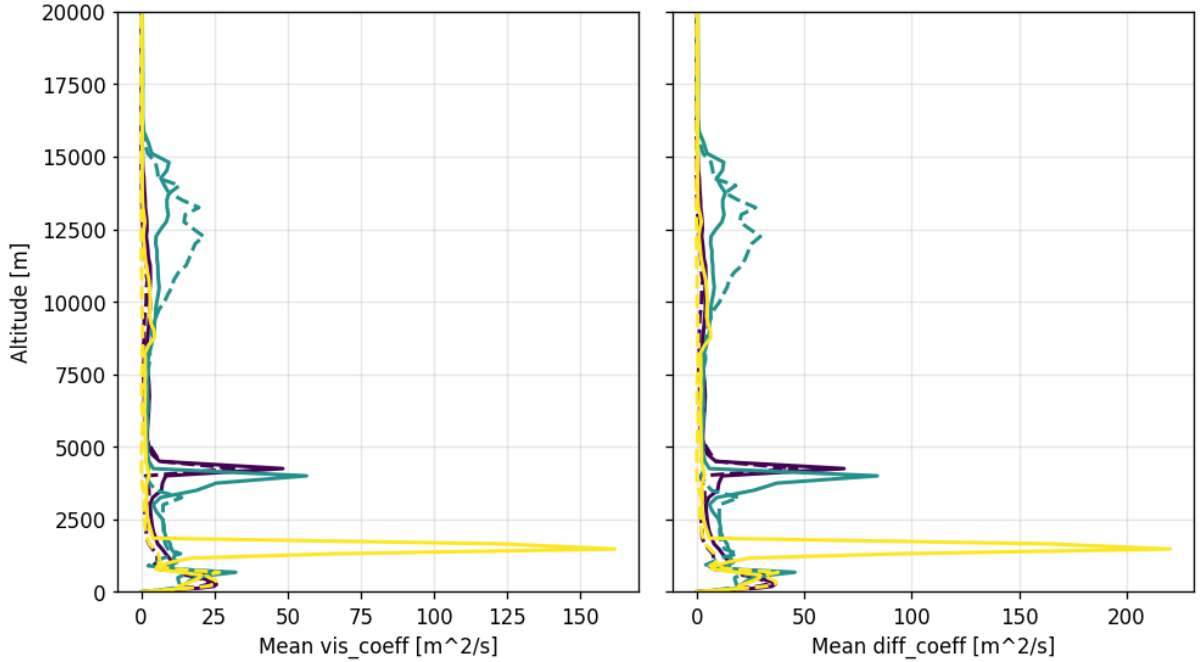
6. Beyond Offline Evaluations: Results after the online integration of the ML models inside MONC using FTorch

6.1. (Online) Ensemble Simulation results: Time series evolution of key bulk diagnostics (+/- IQR colour bands for N>=3 members)



6.2. Domain averaged vertical profile evolution across simulation time: ML driven vs Physics (Smag) driven



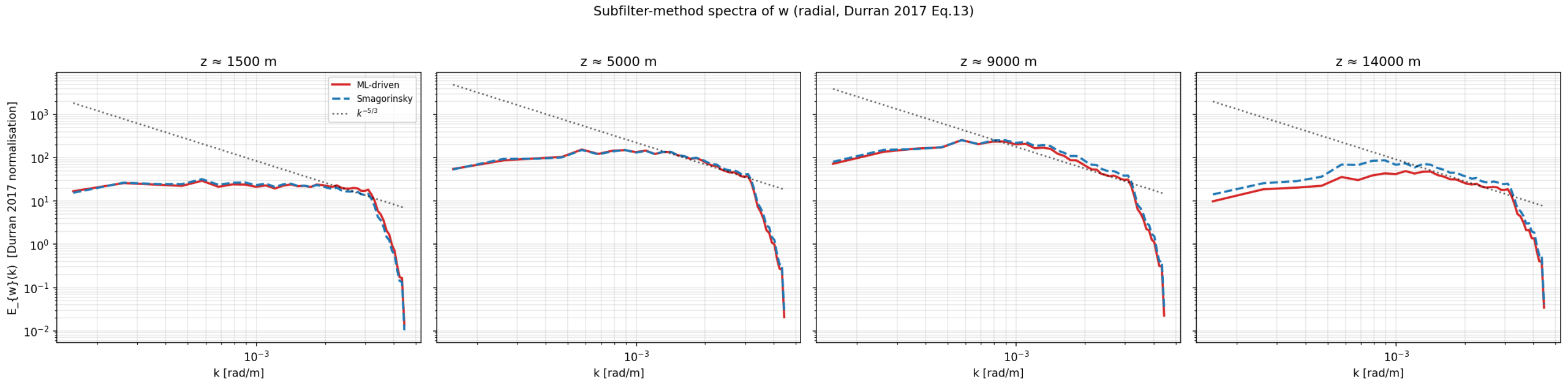


Initial findings seem promising!

- **Stable simulations**: no NaN/blow-up over a full 48 h diurnal cycle (and beyond!). Also confirmed with ensemble simulations.

- **Realistic**: ML (Ri-MLP) tracks Smagorinsky across every bulk diagnostic and at every profile snapshot.

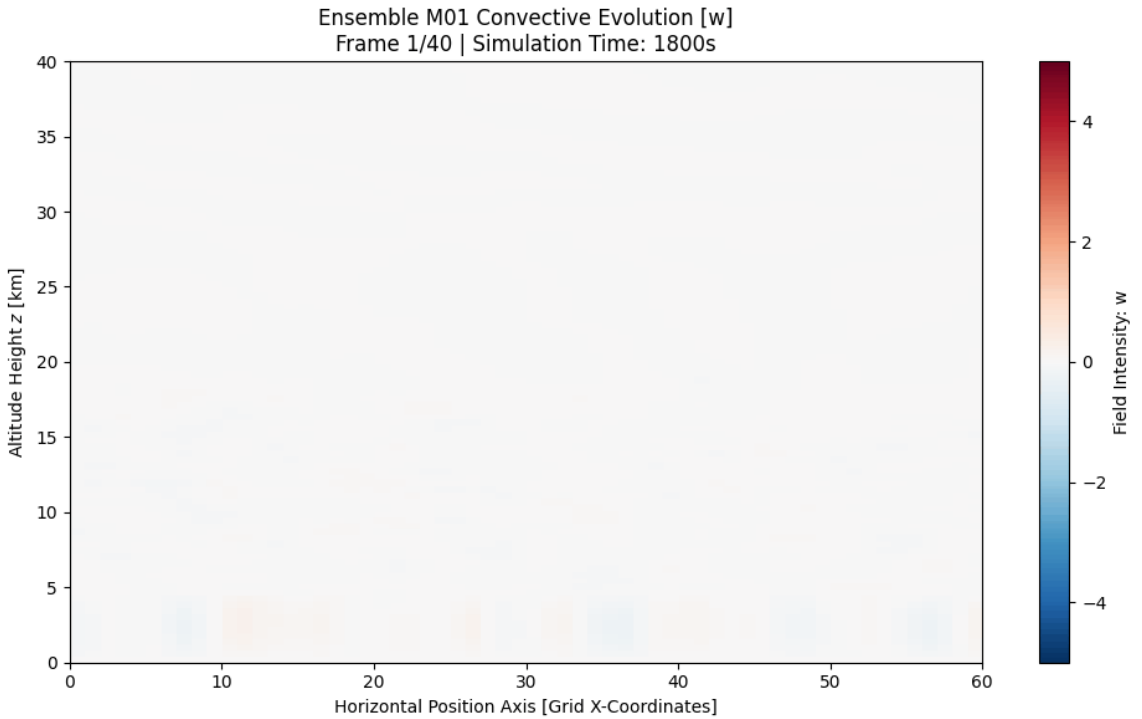
6.3. Power Spectrum analysis for ‘w (vertical vel.)’ across simulation time: ML driven vs Physics (Smag) driven



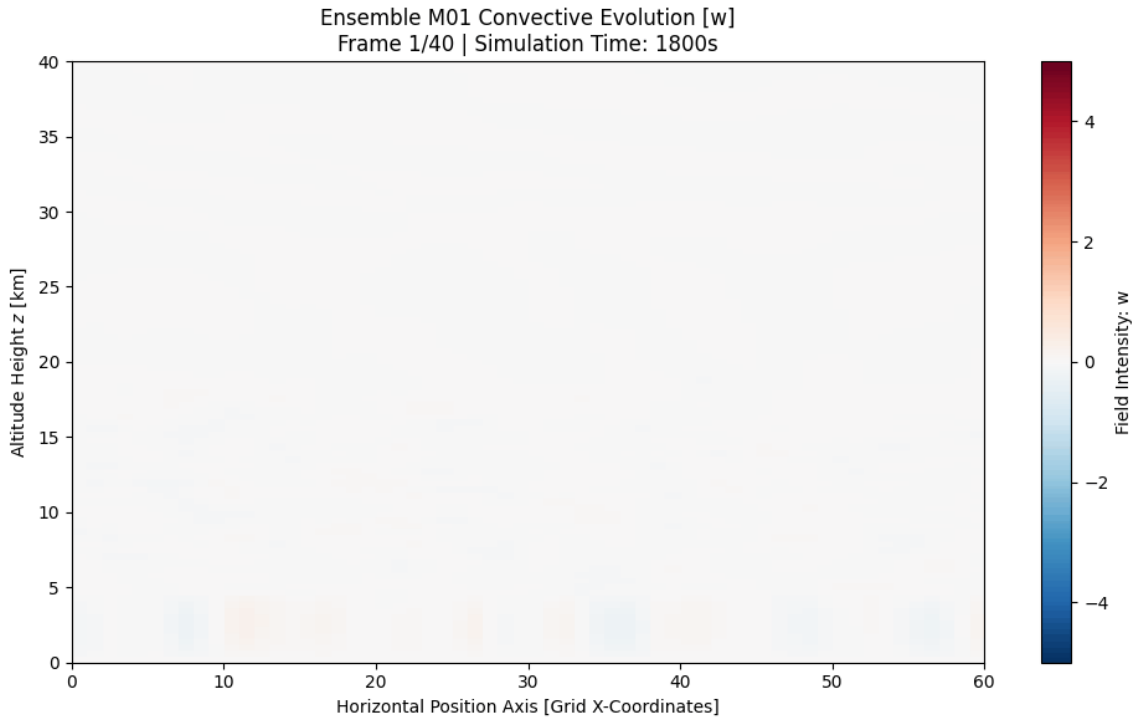
From ~450 hours of model simulation

6.4. GIF Animation of vertical velocity (w) over X-Z plane for a single ensemble member: ML driven vs Physics (Smag) driven simulations for RCE

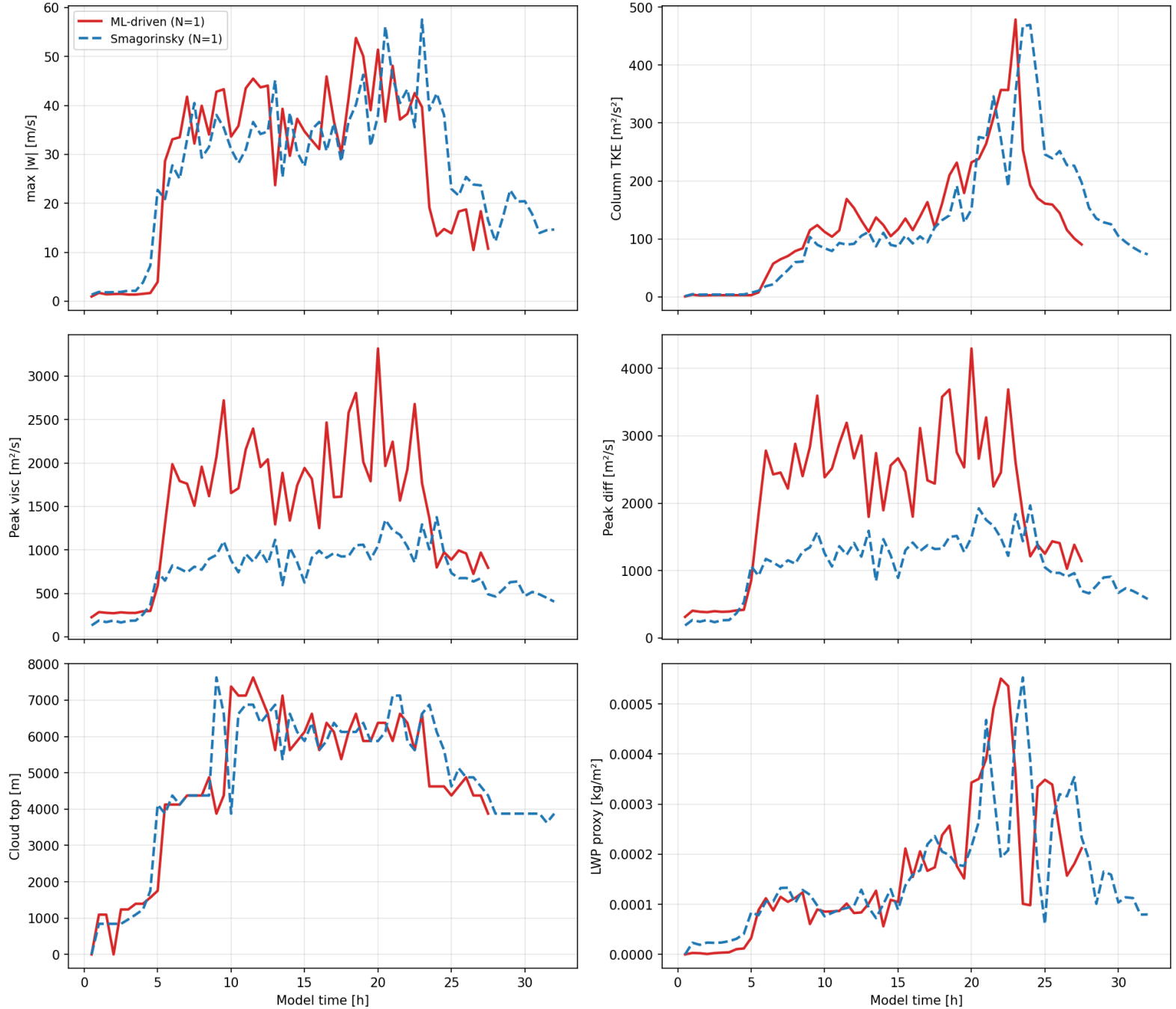
ML driven: m01



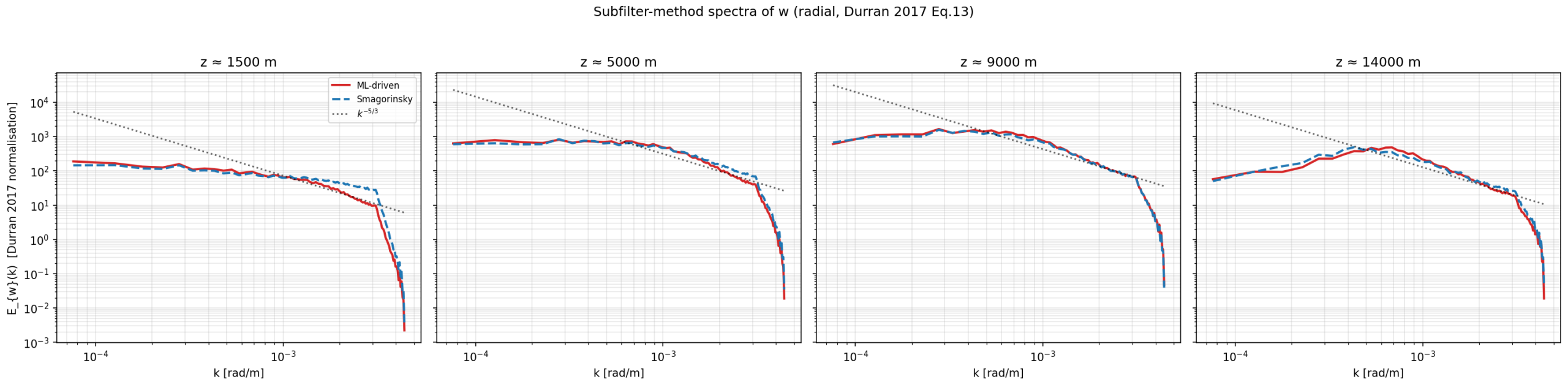
SMAG driven: m01



6.5. (Online) Simulation results (dx=dy=500m, RCE): Time series evolution of key bulk diagnostics



6.6. Power Spectrum analysis for ‘w (vertical vel.)’ across simulation time (dx=dy=500m, RCE): ML driven vs Physics (Smag) driven



Even at a new (unseen) resolution, the ML-SGS driven MONC model performs realistic simulations, while decently emulating the Smagorinsky behaviour!

5. Conclusions

❖ **RQ1: Multi-resolution training from sparse points? →Yes, and it reconstructs full 3D volumes.**

- One model spans a 20× grid resolution gap from 4.05 M sparse points (~1% RCE domain / ~0.2% ARM domain), reconstructing 222.6 M full-domain predictions.
- Now runs stably and realistically online after integration with MONC as well. **Even on unseen resolutions!**

❖ **RQ2: Physics-guided vs. data-driven, at equal data? → Yes: targeted physics wins at equal training data.**

- Explicit Ri-conditioning on a simple MLP gives 158% improvement for active-turbulence viscosity R^2 scores (for ARM) over the data-driven end member (Q4 config).

❖ **RQ3: Should Physics persist at inference, or train-time only? → Physics should persist at inference; conditioning beats gradients-only approach.**

- Architectural Ri-conditioning (Ri-MLP) leads under distribution shift; the auxiliary-only baseline trails.
- But conditioning must be selective, not universal: Q1 config conditions every head, cascades Ri errors, and collapses (R^2 drops to 0.119).

❖ **RQ4: Is MTL the right choice, and do CV weightings transfer? → MTL yes; weighting transfers only partly.**

- Auxiliary Ri prediction improves R^2 by 4.4% (Viscosity) and 13.8% (Diffusivity) as compared to Q4 data-driven config.
- Uncertainty weighting wins by 20–30%; DWA transfers from CV poorly as it oscillates and collapses on deeper neural nets.

❖ **RQ5: Does architectural complexity help or hinder? → Complexity hinders; simplicity generalizes (with same training data).**

- All architectures match on RCE ($R^2 \approx 0.78$); only the MLP/Ri-MLP survive the ARM test.
- ResMLP and TabTransformer show variance explosion (VR up to 5.3×, negative R^2) under distribution shift.

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- Met Office, UK: For CASE funding and HPC support
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- AFESP Office, UoR: For Project funding, Travel and overall support

Thank You

Zenodo Repo Link: PASC 2026

GitHub Repo Link: PASC 2026

Zenodo Repo Link: WCCI 2026

GitHub Repo Link: WCCI 2026

Questions/Feedback/Suggestions/Comments are most welcome!!!

Email: **s.panda@pgr.reading.ac.uk**

Backup slides

Timestep ‘n’
(For each MPI rank, for each column)

